



Does ResNet Learn Good General Purpose Features?

Yang Li, Yafei Zhang, Yulong Xu, Zhuang Miao, Hang Li

College of Command Information Systems

PLA University of Science and Technology

Nanjing, China 210007

solarleeon@outlook.com

ABSTRACT

ResNet with hundreds or even thousands of layers has become the most successful image recognition model in the computer vision community. However, we do not know whether this deeper architecture has better transferability than the traditional models. In this paper, we systematically investigate the well-known ResNet features for different computer vision tasks without fine-tuning. The experimental results show that ResNet did not learn good general purpose features in image retrieval and visual object tracking tasks. GoogLeNet high-level features and VGGNet middle-level features achieve better performance for transfer learning. Therefore, it is not a good idea to choose ResNet model in other computer vision tasks without fine-tuning.

CCS CONCEPTS

•Computing methodologies →Transfer learning;

KEYWORDS

residual network; transfer learning; image retrieval; visual object tracking

1 INTRODUCTION

Convolutional neural networks (CNNs) have become the most successful machine learning approach in the computer vision community since AlexNet [1]. With the depth increasing in network structure, CNNs can achieve better performance on ImageNet [2] classification task. AlexNet contains 8 learned layers [1], VGGNet consists of 19 learned layers [3], GoogLeNet contains 22 learned layers [4], and ResNet [5] has achieved 152 learned layers.

In addition, pre-trained CNN model can also be used in a wide variety of computer vision problems, such as object detection [6], image retrieval [7], object tracking [8] and others [9]. However, if we want to use a pre-trained CNN model to solve other computer vision tasks, we always need a time consuming fine-tuning process. And when we try to use a pre-trained CNN model without fine-tuning in other computer vision tasks, it is natural to ask: whether the deeper model (e.g. ResNet) is better than the shallow one (e.g. VGGNet)? Which CNN model is the best architecture to learn good general purpose features for a variety of computer vision tasks?

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

AIAC '17, Wuhan, China

© 2017 ACM. 978-1-4503-5231-4/17/04...\$15.00

DOI: <http://dx.doi.org/10.1145/3080845.3080864>

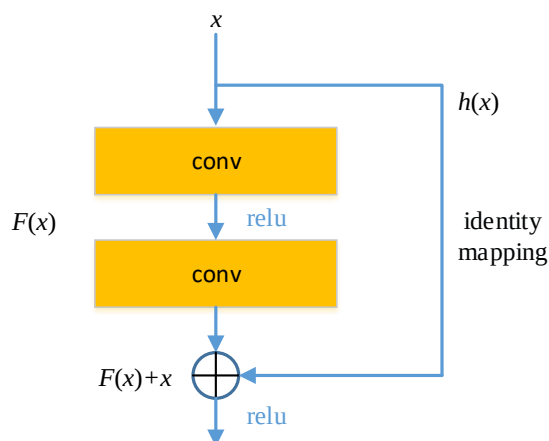


Figure 1: The residual building block.

We are interested in answering these questions because if CNN features are general purpose, we will be able to use them for transfer learning. Furthermore, we believe that the great power of human vision system comes from the feature representation mechanism in the human brain which can be used in many vision tasks. And we want to discover the similar capability in different network architecture.

In this paper, we systematically investigate the well-known ResNet features for different computer vision tasks without fine-tuning. To our slight surprise, ResNet features do not perform well in other computer vision tasks. The following is a summary of our main findings:

(1) The most powerful and deep ResNet image classification model results in a big drop in transfer learning performance. In addition, the transferability of features decreases as the depth increasing of the CNN structure.

(2) Transferable features are well learnt in a single pass CNN model (e.g.VGGNet) rather than an ensemble CNN model (e.g. ResNet). In other words, the VGGNet can be viewed as a general purpose deep model and the ResNet concentrates more on image classification. So, if we want to use a pre-trained CNN model for other computer vision tasks without fine-tuning, ResNet is not a good choice.

The remainder of this paper is organized as follows. Section 2 introduces the efficiency and expressibility of ResNet. Section 3 presents the experimental results and comparisons with several

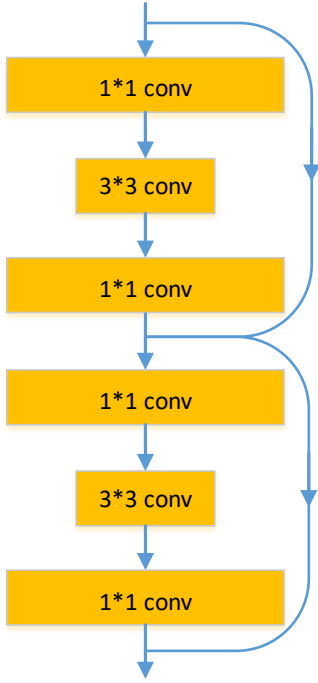


Figure 2: The bottleneck structure of residual units.

state-of-the-art deep network architectures. Section 4 discusses the reason of ResNet high-level features and middle-level features are not good general purpose features. Finally we conclude our paper with future works in Section 5.

2 EFFICIENCY AND EXPRESSIBILITY OF RESNET

The basic idea of ResNet [5] is that residual mapping is easy to optimize. Therefore ResNet proposed using identity shortcut connections to pass residual information across layers without attenuation. The residual building block is shown in Fig. 1. The output of the i -th layer can be defined as:

$$\begin{aligned} y_i &= h(x_i) + F(x_i, W_i) \\ x_{i+1} &= f(y_i) \end{aligned} \quad (1)$$

where x_i and x_{i+1} are input and output of the i -th block, F is a residual mapping function, $h(x_i) = x_i$ is an identity mapping function and f is a ReLU function. He *et al.* [5] also hypothesized that residual framework is more easy to train and lead to more efficient models for the bottleneck designs. Bottleneck structure is illustrated in Fig. 2.

Recently, more and more publications focus on understanding ResNet. Veit *et al.* [10] reveal that the efficiency of ResNet is not because of depth, and residual networks behave just like ensemble of relatively shallow networks. FractalNet [11] demonstrates that residual representation is not fundamental to the success of extremely deep CNNs. ResNet in ResNet (RiR) [12] propose a deep dual stream architecture that incorporates these generalized residual blocks. Wide ResNet [13] increases width and decreases depth

of residual networks by adding more feature planes. Residual networks of Residual networks (RoR) [14] adds level-wise shortcut connections upon original residual networks and promotes the learning capability of residual networks. DenseNet [15] introduces direct connections within all layers of a dense block to improve accuracy.

On the other hand, Yosinski *et al.* [16] experimentally quantify how transferable are features in deep neural networks before ResNet born. Huh *et al.* [17] investigate (only on AlexNet) which aspects of the ImageNet task are most critical for learning general purpose features. However, whether ResNet can learn good general purpose features for other computer vision tasks is still an open problem.

Inspired by these advances, in this work we systematically investigate the transferability of ResNet features. The contribution of this paper includes two aspects. First, we experimentally show ResNet did not learn good general purpose features in image retrieval and visual object tracking tasks. Second, we compare VGGNet features, GoogLeNet features and ResNet features. We find the GoogLeNet high-level features and VGGNet middle-level features perform better for transfer learning. Finally, our results might indicate that researchers have been overestimating the ResNet representation ability. The better performance on image classification makes ResNet loss general purpose representation power.

3 EXPERIMENTS

Our experimental evaluation has two objectives: The first one is to testify the transferability of high-level ResNet features for image retrieval. The second objective is to testify the transferability of middle-level ResNet features for visual object tracking. In order to examine this, we compare to retrieval and tracking results obtained by methods that rely on the VGGNet [3], GoogLeNet [4] and ResNet [5] without fine-tuning. The pre-trained CNN models are available at <http://www.vlfeat.org/matconvnet/pretrained/>. (The model names are imagenet-vgg-verydeep-19, imagenet-googlenet-dag, and imagenet-resnet-152-dag, respectively.)

3.1 Experimental Settings

All implementations are derived from the MatConvNet toolbox [18]. Matlab 2016a is used to conduct all the operations, running on a PC with Intel i5 3.50GHz CPU, 16GB memory.

In our image retrieval experiment, we use the last fully-connected layer features of VGGNet [3], GoogLeNet [4] and ResNet [5] for evaluation with feature normalization. Due to the different architecture of VGGNet, GoogLeNet and ResNet, the dimensions of the feature vector are different. VGGNet features have 4096 dimensions, GoogLeNet features have 1024 dimensions, and ResNet features have 2048 dimensions.

We report the performance on three publicly available image retrieval datasets: Paris6k [19], Oxford5k [20] and Holidays [21]. The Paris6k dataset contains 6392 images, which are collected from Flickr and corresponding to the famous landmarks in Paris. There are 55 hold-out queries evenly distributed over those 11 landmarks are provided as the queries. The Oxford5k dataset includes 5062 images, which are analogously to the previous Paris6k dataset. There are also 55 hold-out queries for evaluation. The Holidays

Table 1: Mean Average Precision performance (%) on three image retrieval datasets.

Methods	Paris6k [19]	Oxford5k [20]	Holidays [21]
VGGNet [3]	60.51	40.63	75.39
GoogLeNet [4]	68.59	46.55	74.19
ResNet [5]	52.66	38.52	62.61

dataset consists of 1491 images of a very large variety of vacation photos from 500 categories, where the first image in each category is used as the query.

To simplify the experiments, the rectangular regions of Paris6k [19] and Oxford5k [20] are ignored, and full images are used for feature extraction. For each query, all the images in the image retrieval datasets are resized into 224*224 and ranked according to their Cosine distances of the feature vectors. Following the standard evaluation protocol, the performance is reported as mean average precision (mAP) using the semantic ground-truth in three datasets.

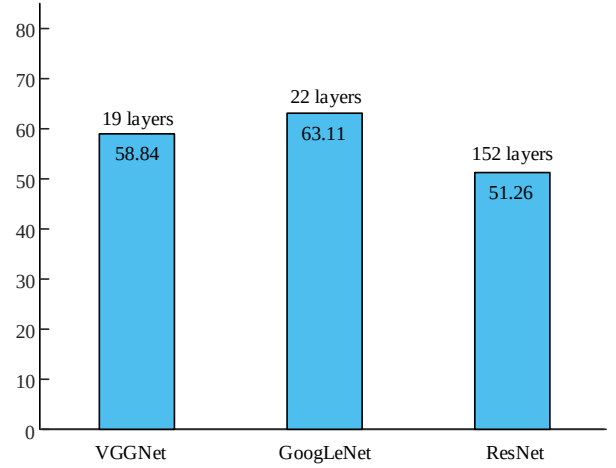
In our visual object tracking experiment, we evaluate the tracking results by using a state-of-the-art deep learning tracking method [8]. All the settings are all the same as [8]. We just replace the feature maps of different CNN models. We adopt VGGNet’s relu1_2, relu2_2, relu3_4, relu4_4, and relu5_4 feature maps, GoogLeNet’s conv1x, norm2, icp2_out, icp7_out, and icp9_out feature maps and ResNet’s conv1xxx, res2cx, res3b7x, res4b35x, and res5cx feature maps for visual tracking task. The name of feature maps are the same name as in the networks, and they are all in the turning point of different resolutions. We evaluate the tracking performance on all 51 sequences from the Online Tracking Benchmark (OTB) dataset [22]. The sequences in this dataset involve various tracking challenges, such as illumination variation, deformation, motion blur, background clutter, etc.

The experimental results are evaluated using overlap precision (OP), distance precision (DP), and center location error (CLE). OP is the percentage of frames in a video where the bounding box overlap is greater than a certain threshold $t \in [0, 1]$. And we present the OP results at a threshold of 0.5. DP is defined as the relative number of frames in the sequence where the CLE is smaller than a certain threshold. And we present the DP values at a threshold of 20 pixels. CLE is the average Euclidean distance between the estimated center location of the target and the ground-truth.

3.2 Experimental Evaluation

3.2.1 Image Retrieval. We start by verifying the image retrieval ability of ResNet high-level features. As can be seen in Table 1, GoogLeNet high-level features achieve the best result with 68.59% and 46.55% on Paris6k and Oxford5k datasets, respectively. VGGNet high-level features achieve the best result with 75.39% on Holidays dataset. ResNet high-level features produce the worst results on three image retrieval datasets. These results indicate that ResNet high-level features are not sufficient to learn general purpose high-level features that are also good for image retrieval.

In addition, we average the retrieval accuracy on three image retrieval datasets in Fig. 3. And we can see that the relative shallow

**Figure 3: Image retrieval performance on different depth models. The results show that features learnt by pre-trained ResNet lead to relative poor result.****Table 2: Comparisons different features on the 51 benchmark sequences.**

Methods	Mean OP (%)	Mean DP (%)	Mean CLE
VGGNet [3]	83.8	89.5	15.6
GoogLeNet [4]	30.6	43.3	43.0
ResNet [5]	48.7	61.5	34.1

VGGNet (19 layers) and GoogLeNet (22 layers) can achieve better image retrieval results than the deeper ResNet (152 layers).

3.2.2 Visual Object Tracking. In Table 2, we present a comparison results on the OTB dataset using mean OP, mean DP and mean CLE over all the 51 sequences [22]. According to the presented results, it can be found that the VGGNet [3] tracker achieves the best result with a mean OP of 83.8%, significantly outperforming ResNet tracker result 48.7%. Meanwhile, the ResNet [5] method provides the results with a mean DP of 61.5% and a mean CLE of 34.1 pixels. GoogLeNet’s middle-level features produce the worst results on visual object tracking.

In addition, we report the results in one-pass evaluation (OPE) using the overlap precision rate over all the 51 videos. The trackers are ranked by the area under the curve (AUC) for success rate plot and center location error at 20 pixels for precision plot in Fig. 4 and Fig. 5. Among the compared methods, the VGGNet [3] provides the best results and obtain AUC score of 65.6%. ResNet tracker only achieves the result of 44.4% AUC score. In summary, the experimental results clearly demonstrate that ResNet middle-level features are not better than shallow networks for visual object tracking.

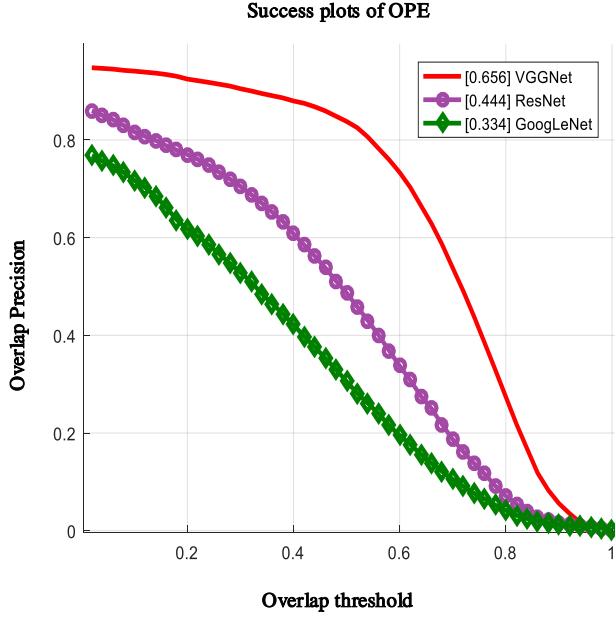


Figure 4: Success plots of overall performance comparison for the 51 videos sequences. The AUC scores of each tracker are used to rank the results as shown in the legend.

4 DISCUSSIONS

From the above experimental results, we can see that the ResNet high-level features and middle-level features are not good general purpose features for other computer vision tasks. And the reason may come from two aspects: First, the ResNet features are overfitting for classification task which makes a big drop in other computer vision tasks. Second, ensemble model is hard for transfer learning because it needs to transfer several sub-models at the same time.

In this paper, we only test ResNet high-level features and middle-level features transferability without fine-tuning. And it has been justified that ResNet can also achieve better results for object detection after fine-tuning [5]. However, we believe that fine-tuning is time consuming. Transferability without fine-tuning goes a step further than previous works and supports more conclusive answers to the quest for good general purpose features.

5 CONCLUSION

In this paper, we systematically investigate the well-known ResNet high-level features and middle-level features for image retrieval and visual object tracking, respectively. To our slight surprise, both high-level and middle-level ResNet features do not perform well in other computer vision tasks without fine-tuning. Furthermore, we show that the transferability of CNN model is negatively affected by depth. Therefore, it is not a good idea to choose the deeper model in other computer vision tasks without fine-tuning.

Deeper model can produce better results on one specific task, but fails to learn good general purpose features. However, how the

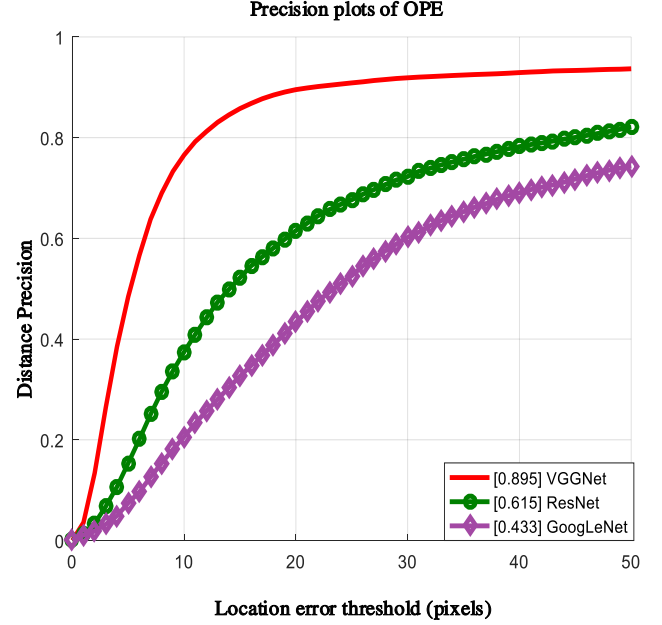


Figure 5: Precision plots of overall performance comparison for the 51 videos sequences. The Mean DP score at 20 pixels are used to rank the results as shown in the legend.

human brain to solve multiple vision tasks at the same time is still an open problem. We hope that this work will pique our colleagues' curiosity and facilitate further research on this fascinating topic. If we can train a CNN model that tackles a broad set of computer vision problems in a unified architecture, it will benefit a lot for us to understand the human vision system.

REFERENCES

- [1] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," *Proc. Advances in Neural Information Processing Systems*, vol. 25, no. 2, pp. 1097-1105, 2012.
- [2] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, and L. Fei-Fei, "ImageNet large scale visual recognition challenge," *International Journal of Computer Vision (IJCV)*, vol. 115, no. 3, pp. 211-252, 2015.
- [3] K. Simonyan, and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *Proc. ICLR*, pp. 1-13, Apr. 2015.
- [4] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich, "Going deeper with convolutions," *Proc. IEEE CVPR*, Boston, MA, USA, pp. 1-9, Jun. 2015.
- [5] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," *Proc. IEEE CVPR*, Las Vegas, Nevada, USA, pp. 770-778, Jun. 2016.
- [6] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," *Proc. IEEE CVPR*, Las Vegas, Nevada, USA, pp. 779-788, Jun. 2016.
- [7] A. Gordo, J. Almazan, J. Revaud, and D. Larlus, "Deep Image Retrieval: Learning global representations for image search," *arXiv preprint, arXiv:1604.01325*, 2016.
- [8] Y. Li, Y. Zhang, Y. Xu, J. Wang, and Z. Miao, "Robust scale adaptive kernel correlation filter tracker with hierarchical convolutional features," *IEEE Signal Processing Letters*, vol. 23, no. 8, pp. 1136-1140, Aug. 2016.
- [9] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436-444, May. 2015.
- [10] A. Veit, M. Wilber, and S. Belongie, "Residual networks are exponential ensembles of relatively shallow networks," *arXiv preprint, arXiv:1605.06431*, 2016.

- [11] G. Larsson, M. Maire, and G. Shakhnarovich, "FractalNet: Ultra-deep neural networks without residuals," arXiv preprint, arXiv:1605.07648, 2016.
- [12] S. Targ, D. Almeida, and K. Lyman, "Resnet in Resnet: Generalizing residual architectures," arXiv preprint, arXiv:1603.08029, 2016.
- [13] S. Zagoruyko, and N. Komodakis, "Wide residual networks," arXiv preprint, arXiv:1605.07146, 2016.
- [14] K. Zhang, M. Sun, T. X. Han, X. Yuan, L. Guo, and T. Liu, "Residual networks of residual networks: Multilevel residual networks," arXiv preprint, arXiv:1608.02908, 2016.
- [15] G. Huang, Z. Liu, and K. Q. Weinberger, "Densely connected convolutional networks," arXiv preprint, arXiv:1608.06993, 2016.
- [16] J. Yosinski, J. Clune, Y. Bengio, and H. Lipson, "How transferable are features in deep neural networks," Proc. Advances in Neural Information Processing Systems, pp. 3320-3328, 2014.
- [17] M. Huh, P. Agrawal, and A. A. Efros, "What makes ImageNet good for transfer learning," arXiv preprint, arXiv:1608.08614, 2016.
- [18] A. Vedaldi and K. Lenc, "MatConvNet: Convolutional neural networks for MATLAB," Proc. 23th Int. Conf. Multimed., pp. 689-692, Oct. 2015.
- [19] J. Philbin, O. Chum, M. Isard, J. Sivic, and A. Zisserman, "Lost in quantization: Improving particular object retrieval in large scale image databases," Proc. IEEE CVPR, Anchorage, Alaska, USA, pp. 1-8, Jun. 2008.
- [20] J. Philbin, O. Chum, M. Isard, J. Sivic, and A. Zisserman, "Object retrieval with large vocabularies and fast spatial matching," Proc. IEEE CVPR, pp. 1-8, Jun. 2007.
- [21] H. Jegou, M. Douze, and C. Schmid, "Hamming embedding and weak geometry consistency for large scale image search," Proc. IEEE ECCV, Marseille, France, pp. 304-317, Oct. 2008.
- [22] Y. Wu, J. Lim, and M.-H. Yang, "Online object tracking: A benchmark," Proc. IEEE CVPR, Portland, Oregon, USA, pp. 2411-2418, Jun. 2013.